iplpredictor

February 20, 2026

[2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split,GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.metrics import␣
↪classification_report,f1_score,ConfusionMatrixDisplay,accuracy_score
```

[3]:
```
data_df = pd.read_csv('E:\\PythonCertificates\\IPL prediction application\\ipl.
↪csv')
```

[4]:
```
data_df.head(5)
```

[4]:
```
                        team1                        team2  match_date  \
0  Royal Challengers Bangalore        Kolkata Knight Riders  2008-04-18
1              Kings XI Punjab          Chennai Super Kings  2008-04-19
2             Delhi Daredevils             Rajasthan Royals  2008-04-19
3               Mumbai Indians  Royal Challengers Bangalore  2008-04-20
4        Kolkata Knight Riders              Deccan Chargers  2008-04-20

                   toss_winner toss_decision                      winner  \
0  Royal Challengers Bangalore         field        Kolkata Knight Riders
1          Chennai Super Kings           bat          Chennai Super Kings
2             Rajasthan Royals           bat             Delhi Daredevils
3               Mumbai Indians           bat  Royal Challengers Bangalore
4              Deccan Chargers           bat        Kolkata Knight Riders

  player_of_match                                       venue        city  \
0     BB McCullum                        M Chinnaswamy Stadium   Bangalore
1      MEK Hussey  Punjab Cricket Association Stadium, Mohali  Chandigarh
2     MF Maharoof                              Feroz Shah Kotla       Delhi
3      MV Boucher                              Wankhede Stadium      Mumbai
4       DJ Hussey                                  Eden Gardens     Kolkata
```

```
                                        team1_players  \
0  R Dravid, W Jaffer, V Kohli, JH Kallis, CL Whi…
1  K Goel, JR Hopes, KC Sangakkara, Yuvraj Singh,…
2  G Gambhir, V Sehwag, S Dhawan, MK Tiwary, KD K…
3  L Ronchi, ST Jayasuriya, DJ Thornely, RV Uthap…
4  WP Saha, BB McCullum, RT Ponting, SC Ganguly, …

                                        team2_players  season  match_number  \
0  SC Ganguly, BB McCullum, RT Ponting, DJ Hussey…       1             1
1  PA Patel, ML Hayden, MEK Hussey, MS Dhoni, SK …       1             2
2  T Kohli, YK Pathan, SR Watson, M Kaif, DS Lehm…       1             3
3  S Chanderpaul, R Dravid, LRPL Taylor, JH Kalli…       1             4
4  AC Gilchrist, Y Venugopal Rao, VVS Laxman, A S…       1             5

  match_type   result  result_margin  target_runs  target_overs super_over
0     League     runs          140.0        223.0          20.0          N
1     League     runs           33.0        241.0          20.0          N
2     League  wickets            9.0        130.0          20.0          N
3     League  wickets            5.0        166.0          20.0          N
4     League  wickets            5.0        111.0          20.0          N
```

[5]:
```
data_df.shape
```

[5]:
```
(1169, 19)
```

[6]:
```
data_df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1169 entries, 0 to 1168
Data columns (total 19 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   team1            1169 non-null   str
 1   team2            1169 non-null   str
 2   match_date       1169 non-null   str
 3   toss_winner      1169 non-null   str
 4   toss_decision    1169 non-null   str
 5   winner           1146 non-null   str
 6   player_of_match  1169 non-null   str
 7   venue            1169 non-null   str
 8   city             1118 non-null   str
 9   team1_players    1169 non-null   str
 10  team2_players    1169 non-null   str
 11  season           1169 non-null   int64
 12  match_number     1169 non-null   int64
 13  match_type       1169 non-null   str
 14  result           1169 non-null   str
 15  result_margin    1146 non-null   float64
```

```
 16  target_runs      1163 non-null   float64
 17  target_overs     1163 non-null   float64
 18  super_over       1169 non-null   str
dtypes: float64(3), int64(2), str(14)
memory usage: 173.7 KB
```

[7]:
```python
data_df.describe()
```

[7]:

	season	match_number	result_margin	target_runs	target_overs
count	1169.000000	1169.000000	1146.000000	1163.000000	1163.000000
mean	9.687767	33.359281	17.364747	167.253654	19.768014
std	5.253100	19.430749	21.919255	34.222771	1.542583
min	1.000000	1.000000	1.000000	43.000000	5.000000
25%	5.000000	17.000000	6.000000	147.000000	20.000000
50%	10.000000	33.000000	8.000000	167.000000	20.000000
75%	15.000000	49.000000	20.000000	188.000000	20.000000
max	18.000000	76.000000	146.000000	288.000000	20.000000

[8]:
```python
data_df.isnull().sum().sort_values(ascending=False)
```

[8]:
```
city               51
winner             23
result_margin      23
target_overs        6
target_runs         6
team1               0
team2               0
match_date          0
player_of_match     0
toss_decision       0
toss_winner         0
team2_players       0
team1_players       0
venue               0
season              0
result              0
match_type          0
match_number        0
super_over          0
dtype: int64
```

[9]:
```python
data_df.replace({
    'Delhi Daredevils': 'Delhi Capitals',
    'Kings XI Punjab': 'Punjab Kings',
    'Royal Challengers Bangalore': 'Royal Challengers Bengaluru',
    'Rising Pune Supergiants': 'Rising Pune Supergiant'
}, inplace=True)
```

[9]:

```
                          team1                       team2  match_date  \
0     Royal Challengers Bengaluru        Kolkata Knight Riders  2008-04-18
1                    Punjab Kings          Chennai Super Kings  2008-04-19
2                  Delhi Capitals             Rajasthan Royals  2008-04-19
3                  Mumbai Indians  Royal Challengers Bengaluru  2008-04-20
4           Kolkata Knight Riders              Deccan Chargers  2008-04-20
…                               …                           …           …
1164         Lucknow Super Giants  Royal Challengers Bengaluru  2025-05-27
1165                 Punjab Kings  Royal Challengers Bengaluru  2025-05-29
1166               Mumbai Indians               Gujarat Titans  2025-05-30
1167               Mumbai Indians                 Punjab Kings  2025-06-01
1168  Royal Challengers Bengaluru                 Punjab Kings  2025-06-03

                      toss_winner toss_decision                       winner  \
0     Royal Challengers Bengaluru         field        Kolkata Knight Riders
1             Chennai Super Kings           bat          Chennai Super Kings
2                Rajasthan Royals           bat                Delhi Capitals
3                  Mumbai Indians           bat  Royal Challengers Bengaluru
4                 Deccan Chargers           bat        Kolkata Knight Riders
…                               …           …                           …
1164  Royal Challengers Bengaluru         field  Royal Challengers Bengaluru
1165  Royal Challengers Bengaluru         field  Royal Challengers Bengaluru
1166               Mumbai Indians           bat                Mumbai Indians
1167                 Punjab Kings         field                  Punjab Kings
1168                 Punjab Kings         field  Royal Challengers Bengaluru

     player_of_match                                            venue  \
0        BB McCullum                              M Chinnaswamy Stadium
1         MEK Hussey         Punjab Cricket Association Stadium, Mohali
2        MF Maharoof                                   Feroz Shah Kotla
3         MV Boucher                                   Wankhede Stadium
4          DJ Hussey                                       Eden Gardens
…                …                                                   …
1164       JM Sharma  Bharat Ratna Shri Atal Bihari Vajpayee Ekana C…
1165   Suyash Sharma  Maharaja Yadavindra Singh International Cricke…
1166       RG Sharma  Maharaja Yadavindra Singh International Cricke…
1167         SS Iyer                   Narendra Modi Stadium, Ahmedabad
1168       KH Pandya                   Narendra Modi Stadium, Ahmedabad

                city                                       team1_players  \
0          Bangalore  R Dravid, W Jaffer, V Kohli, JH Kallis, CL Whi…
1         Chandigarh  K Goel, JR Hopes, KC Sangakkara, Yuvraj Singh,…
2              Delhi  G Gambhir, V Sehwag, S Dhawan, MK Tiwary, KD K…
3             Mumbai  L Ronchi, ST Jayasuriya, DJ Thornely, RV Uthap…
4            Kolkata  WP Saha, BB McCullum, RT Ponting, SC Ganguly, …
…                …                                                  …
1164         Lucknow  Akash Singh, MR Marsh, MP Breetzke, RR Pant, N…
```

```
1165  New Chandigarh  Priyansh Arya, P Simran Singh, JP Inglis, SS I…
1166  New Chandigarh  Ashwani Kumar, RG Sharma, JM Bairstow, SA Yada…
1167       Ahmedabad  Ashwani Kumar, RG Sharma, JM Bairstow, Tilak V…
1168       Ahmedabad  Suyash Sharma, PD Salt, V Kohli, MA Agarwal, R…

                                          team2_players  season  match_number  \
0     SC Ganguly, BB McCullum, RT Ponting, DJ Hussey…       1             1
1     PA Patel, ML Hayden, MEK Hussey, MS Dhoni, SK …       1             2
2     T Kohli, YK Pathan, SR Watson, M Kaif, DS Lehm…       1             3
3     S Chanderpaul, R Dravid, LRPL Taylor, JH Kalli…       1             4
4     AC Gilchrist, Y Venugopal Rao, VVS Laxman, A S…       1             5
…                                                   …     …             …
1164  Suyash Sharma, PD Salt, V Kohli, RM Patidar, L…      18            70
1165  Suyash Sharma, PD Salt, V Kohli, MA Agarwal, R…      18            71
1166  Mohammed Siraj, B Sai Sudharsan, Shubman Gill,…      18            72
1167  YS Chahal, Priyansh Arya, P Simran Singh, JP I…      18            73
1168  YS Chahal, Priyansh Arya, P Simran Singh, JP I…      18            74

       match_type   result  result_margin  target_runs  target_overs  \
0          League     runs          140.0        223.0          20.0
1          League     runs           33.0        241.0          20.0
2          League  wickets            9.0        130.0          20.0
3          League  wickets            5.0        166.0          20.0
4          League  wickets            5.0        111.0          20.0
…           …      …            …           …            …
1164       League  wickets            6.0        228.0          20.0
1165  Qualifier 1  wickets            8.0        102.0          20.0
1166   Eliminator     runs           20.0        229.0          20.0
1167  Qualifier 2  wickets            5.0        204.0          20.0
1168        Final     runs            6.0        191.0          20.0

     super_over
0             N
1             N
2             N
3             N
4             N
…         …
1164          N
1165          N
1166          N
1167          N
1168          N

[1169 rows x 19 columns]
```

[10]:
```
data_df = data_df[data_df['winner'].notna()]
```

[11]:
```
data_df['winner'].value_counts()
```

[11]:
```
winner
Mumbai Indians                 151
Chennai Super Kings            142
Kolkata Knight Riders          135
Royal Challengers Bengaluru    132
Punjab Kings                   119
Delhi Capitals                 118
Rajasthan Royals               114
Sunrisers Hyderabad             93
Gujarat Titans                  37
Lucknow Super Giants            30
Deccan Chargers                 29
Rising Pune Supergiant          15
Gujarat Lions                   13
Pune Warriors                   12
Kochi Tuskers Kerala             6
Name: count, dtype: int64
```

[12]:
```
team1_wins = data_df[data_df['winner'] == data_df['team1']].shape[0]
team2_wins = data_df[data_df['winner'] == data_df['team2']].shape[0]

print("Team1 Wins:", team1_wins)
print("Team2 Wins:", team2_wins)
```

```
Team1 Wins: 581
Team2 Wins: 565
```

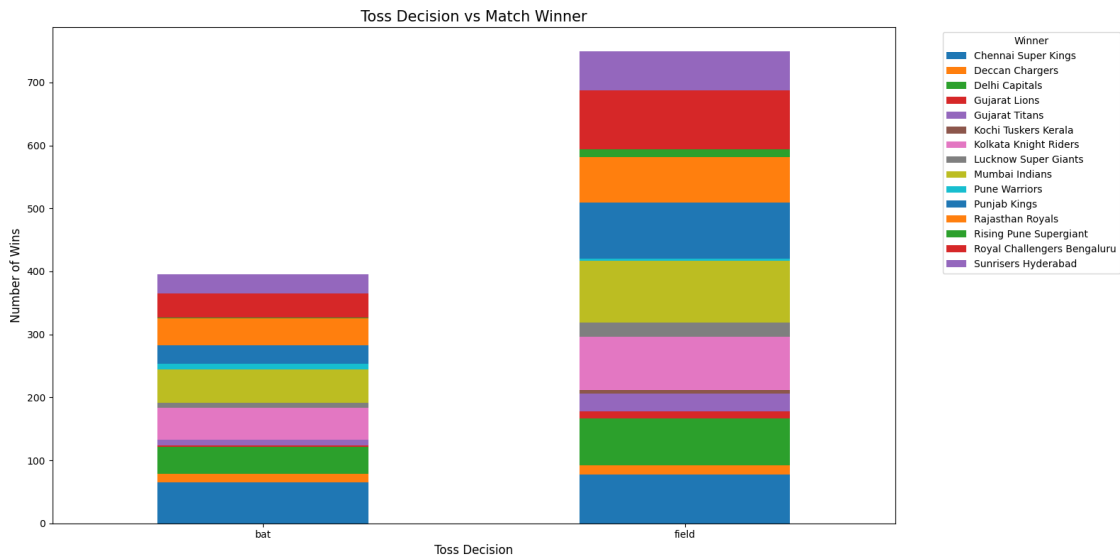
[13]:
```
toss_win_match_win = data_df[data_df['toss_winner'] == data_df['winner']].
 ↪shape[0]
total_matches = data_df.shape[0]

print("Toss winner also won match %:",
      toss_win_match_win/total_matches * 100)
```

```
Toss winner also won match %: 51.57068062827225
```

[14]:
```
plt.figure(figsize=(16, 8))

ax = pd.crosstab(data_df['toss_decision'], data_df['winner']).plot(
    kind='bar',
    stacked=True,
    figsize=(16, 8)
)

plt.title("Toss Decision vs Match Winner", fontsize=15)
plt.xlabel("Toss Decision", fontsize=12)
```

```
plt.ylabel("Number of Wins", fontsize=12)
plt.xticks(rotation=0)
plt.legend(title='Winner', bbox_to_anchor=(1.05, 1), loc='upper left')

plt.tight_layout()
plt.show()
```

```
<Figure size 1600x800 with 0 Axes>
```



[15]:
```
pd.crosstab(data_df['venue'], data_df['winner'])
```

[15]:
```
winner                                          Chennai Super Kings  \
venue
Arun Jaitley Stadium                                              1
Arun Jaitley Stadium, Delhi                                       2
Barabati Stadium                                                  0
Barsapara Cricket Stadium, Guwahati                               0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                1
Brabourne Stadium                                                 1
Brabourne Stadium, Mumbai                                         0
Buffalo Park                                                      1
De Beers Diamond Oval                                             1
Dr DY Patil Sports Academy                                        3
Dr DY Patil Sports Academy, Mumbai                                3
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                0
Dubai International Cricket Stadium                               9
Eden Gardens                                                      5
Eden Gardens, Kolkata                                             2
```

```
Feroz Shah Kotla                                                    5
Green Park                                                          0
Himachal Pradesh Cricket Association Stadium                        1
Himachal Pradesh Cricket Association Stadium, D…                  1
Holkar Cricket Stadium                                              0
JSCA International Stadium Complex                                  3
Kingsmead                                                           1
M Chinnaswamy Stadium                                               4
M Chinnaswamy Stadium, Bengaluru                                    1
M.Chinnaswamy Stadium                                               1
MA Chidambaram Stadium                                              7
MA Chidambaram Stadium, Chepauk                                    33
MA Chidambaram Stadium, Chepauk, Chennai                           11
Maharaja Yadavindra Singh International Cricket…                  0
Maharaja Yadavindra Singh International Cricket…                  0
Maharashtra Cricket Association Stadium                             5
Maharashtra Cricket Association Stadium, Pune                       1
Narendra Modi Stadium, Ahmedabad                                    2
Nehru Stadium                                                       0
New Wanderers Stadium                                               1
Newlands                                                            0
OUTsurance Oval                                                     0
Punjab Cricket Association IS Bindra Stadium                        0
Punjab Cricket Association IS Bindra Stadium, M…                  0
Punjab Cricket Association IS Bindra Stadium, M…                  0
Punjab Cricket Association Stadium, Mohali                          3
Rajiv Gandhi International Stadium                                  1
Rajiv Gandhi International Stadium, Uppal                           2
Rajiv Gandhi International Stadium, Uppal, Hyde…                  0
Sardar Patel Stadium, Motera                                        0
Saurashtra Cricket Association Stadium                              0
Sawai Mansingh Stadium                                              3
Sawai Mansingh Stadium, Jaipur                                      0
Shaheed Veer Narayan Singh International Stadium                    0
Sharjah Cricket Stadium                                             3
Sheikh Zayed Stadium                                                3
St George's Park                                                    2
Subrata Roy Sahara Stadium                                          1
SuperSport Park                                                     2
Vidarbha Cricket Association Stadium, Jamtha                        0
Wankhede Stadium                                                    7
Wankhede Stadium, Mumbai                                            6
Zayed Cricket Stadium, Abu Dhabi                                    1

winner                                              Deccan Chargers  \
venue
Arun Jaitley Stadium                                              0
```

Arun Jaitley Stadium, Delhi	0
Barabati Stadium	3
Barsapara Cricket Stadium, Guwahati	0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…	0
Brabourne Stadium	0
Brabourne Stadium, Mumbai	0
Buffalo Park	0
De Beers Diamond Oval	1
Dr DY Patil Sports Academy	2
Dr DY Patil Sports Academy, Mumbai	0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dubai International Cricket Stadium	0
Eden Gardens	0
Eden Gardens, Kolkata	0
Feroz Shah Kotla	2
Green Park	0
Himachal Pradesh Cricket Association Stadium	2
Himachal Pradesh Cricket Association Stadium, D…	0
Holkar Cricket Stadium	0
JSCA International Stadium Complex	0
Kingsmead	2
M Chinnaswamy Stadium	1
M Chinnaswamy Stadium, Bengaluru	0
M.Chinnaswamy Stadium	0
MA Chidambaram Stadium	0
MA Chidambaram Stadium, Chepauk	2
MA Chidambaram Stadium, Chepauk, Chennai	0
Maharaja Yadavindra Singh International Cricket…	0
Maharaja Yadavindra Singh International Cricket…	0
Maharashtra Cricket Association Stadium	0
Maharashtra Cricket Association Stadium, Pune	0
Narendra Modi Stadium, Ahmedabad	0
Nehru Stadium	1
New Wanderers Stadium	2
Newlands	2
OUTsurance Oval	0
Punjab Cricket Association IS Bindra Stadium	0
Punjab Cricket Association IS Bindra Stadium, M…	0
Punjab Cricket Association IS Bindra Stadium, M…	0
Punjab Cricket Association Stadium, Mohali	0
Rajiv Gandhi International Stadium	0
Rajiv Gandhi International Stadium, Uppal	3
Rajiv Gandhi International Stadium, Uppal, Hyde…	0
Sardar Patel Stadium, Motera	0
Saurashtra Cricket Association Stadium	0
Sawai Mansingh Stadium	0

```
Sawai Mansingh Stadium, Jaipur                                    0
Shaheed Veer Narayan Singh International Stadium                  0
Sharjah Cricket Stadium                                           0
Sheikh Zayed Stadium                                              0
St George's Park                                                  0
Subrata Roy Sahara Stadium                                        1
SuperSport Park                                                   2
Vidarbha Cricket Association Stadium, Jamtha                      2
Wankhede Stadium                                                  1
Wankhede Stadium, Mumbai                                          0
Zayed Cricket Stadium, Abu Dhabi                                  0

winner                                              Delhi Capitals  \
venue
Arun Jaitley Stadium                                             7
Arun Jaitley Stadium, Delhi                                      6
Barabati Stadium                                                 0
Barsapara Cricket Stadium, Guwahati                              0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…               2
Brabourne Stadium                                                0
Brabourne Stadium, Mumbai                                        4
Buffalo Park                                                     1
De Beers Diamond Oval                                            0
Dr DY Patil Sports Academy                                       1
Dr DY Patil Sports Academy, Mumbai                               2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…               2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…               3
Dubai International Cricket Stadium                              6
Eden Gardens                                                     2
Eden Gardens, Kolkata                                            0
Feroz Shah Kotla                                                23
Green Park                                                       1
Himachal Pradesh Cricket Association Stadium                     1
Himachal Pradesh Cricket Association Stadium, D…               1
Holkar Cricket Stadium                                           0
JSCA International Stadium Complex                               0
Kingsmead                                                        3
M Chinnaswamy Stadium                                            3
M Chinnaswamy Stadium, Bengaluru                                 1
M.Chinnaswamy Stadium                                            1
MA Chidambaram Stadium                                           0
MA Chidambaram Stadium, Chepauk                                  2
MA Chidambaram Stadium, Chepauk, Chennai                         2
Maharaja Yadavindra Singh International Cricket…               0
Maharaja Yadavindra Singh International Cricket…               0
Maharashtra Cricket Association Stadium                          2
Maharashtra Cricket Association Stadium, Pune                    0
```

```
Narendra Modi Stadium, Ahmedabad                                    4
Nehru Stadium                                                       1
New Wanderers Stadium                                               1
Newlands                                                            1
OUTsurance Oval                                                     1
Punjab Cricket Association IS Bindra Stadium                        0
Punjab Cricket Association IS Bindra Stadium, M…                  0
Punjab Cricket Association IS Bindra Stadium, M…                  0
Punjab Cricket Association Stadium, Mohali                          1
Rajiv Gandhi International Stadium                                  1
Rajiv Gandhi International Stadium, Uppal                           4
Rajiv Gandhi International Stadium, Uppal, Hyde…                  1
Sardar Patel Stadium, Motera                                        1
Saurashtra Cricket Association Stadium                              1
Sawai Mansingh Stadium                                              2
Sawai Mansingh Stadium, Jaipur                                      1
Shaheed Veer Narayan Singh International Stadium                    4
Sharjah Cricket Stadium                                             5
Sheikh Zayed Stadium                                                2
St George's Park                                                    1
Subrata Roy Sahara Stadium                                          1
SuperSport Park                                                     2
Vidarbha Cricket Association Stadium, Jamtha                        0
Wankhede Stadium                                                    3
Wankhede Stadium, Mumbai                                            3
Zayed Cricket Stadium, Abu Dhabi                                    1

winner                                                Gujarat Lions  \
venue
Arun Jaitley Stadium                                                0
Arun Jaitley Stadium, Delhi                                         0
Barabati Stadium                                                    0
Barsapara Cricket Stadium, Guwahati                                 0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                  0
Brabourne Stadium                                                   0
Brabourne Stadium, Mumbai                                           0
Buffalo Park                                                        0
De Beers Diamond Oval                                               0
Dr DY Patil Sports Academy                                          0
Dr DY Patil Sports Academy, Mumbai                                  0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                  0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                  0
Dubai International Cricket Stadium                                 0
Eden Gardens                                                        2
Eden Gardens, Kolkata                                               0
Feroz Shah Kotla                                                    1
Green Park                                                          2
```

```
Himachal Pradesh Cricket Association Stadium                    0
Himachal Pradesh Cricket Association Stadium, D…              0
Holkar Cricket Stadium                                          0
JSCA International Stadium Complex                              0
Kingsmead                                                       0
M Chinnaswamy Stadium                                           1
M Chinnaswamy Stadium, Bengaluru                                0
M.Chinnaswamy Stadium                                           0
MA Chidambaram Stadium                                          0
MA Chidambaram Stadium, Chepauk                                 0
MA Chidambaram Stadium, Chepauk, Chennai                        0
Maharaja Yadavindra Singh International Cricket…              0
Maharaja Yadavindra Singh International Cricket…              0
Maharashtra Cricket Association Stadium                         1
Maharashtra Cricket Association Stadium, Pune                   0
Narendra Modi Stadium, Ahmedabad                                0
Nehru Stadium                                                   0
New Wanderers Stadium                                           0
Newlands                                                        0
OUTsurance Oval                                                 0
Punjab Cricket Association IS Bindra Stadium                    0
Punjab Cricket Association IS Bindra Stadium, M…              2
Punjab Cricket Association IS Bindra Stadium, M…              0
Punjab Cricket Association Stadium, Mohali                      0
Rajiv Gandhi International Stadium                              0
Rajiv Gandhi International Stadium, Uppal                       0
Rajiv Gandhi International Stadium, Uppal, Hyde…              0
Sardar Patel Stadium, Motera                                    0
Saurashtra Cricket Association Stadium                          3
Sawai Mansingh Stadium                                          0
Sawai Mansingh Stadium, Jaipur                                  0
Shaheed Veer Narayan Singh International Stadium                0
Sharjah Cricket Stadium                                         0
Sheikh Zayed Stadium                                            0
St George's Park                                                0
Subrata Roy Sahara Stadium                                      0
SuperSport Park                                                 0
Vidarbha Cricket Association Stadium, Jamtha                    0
Wankhede Stadium                                                1
Wankhede Stadium, Mumbai                                        0
Zayed Cricket Stadium, Abu Dhabi                                0

winner                                                Gujarat Titans  \
venue
Arun Jaitley Stadium                                               0
Arun Jaitley Stadium, Delhi                                        2
Barabati Stadium                                                   0
```

Barsapara Cricket Stadium, Guwahati	0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…	1
Brabourne Stadium	0
Brabourne Stadium, Mumbai	2
Buffalo Park	0
De Beers Diamond Oval	0
Dr DY Patil Sports Academy	0
Dr DY Patil Sports Academy, Mumbai	2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dubai International Cricket Stadium	0
Eden Gardens	0
Eden Gardens, Kolkata	3
Feroz Shah Kotla	0
Green Park	0
Himachal Pradesh Cricket Association Stadium	0
Himachal Pradesh Cricket Association Stadium, D…	0
Holkar Cricket Stadium	0
JSCA International Stadium Complex	0
Kingsmead	0
M Chinnaswamy Stadium	0
M Chinnaswamy Stadium, Bengaluru	2
M.Chinnaswamy Stadium	0
MA Chidambaram Stadium	0
MA Chidambaram Stadium, Chepauk	0
MA Chidambaram Stadium, Chepauk, Chennai	0
Maharaja Yadavindra Singh International Cricket…	1
Maharaja Yadavindra Singh International Cricket…	0
Maharashtra Cricket Association Stadium	0
Maharashtra Cricket Association Stadium, Pune	3
Narendra Modi Stadium, Ahmedabad	13
Nehru Stadium	0
New Wanderers Stadium	0
Newlands	0
OUTsurance Oval	0
Punjab Cricket Association IS Bindra Stadium	0
Punjab Cricket Association IS Bindra Stadium, M…	0
Punjab Cricket Association IS Bindra Stadium, M…	1
Punjab Cricket Association Stadium, Mohali	0
Rajiv Gandhi International Stadium	0
Rajiv Gandhi International Stadium, Uppal	0
Rajiv Gandhi International Stadium, Uppal, Hyde…	1
Sardar Patel Stadium, Motera	0
Saurashtra Cricket Association Stadium	0
Sawai Mansingh Stadium	0
Sawai Mansingh Stadium, Jaipur	2
Shaheed Veer Narayan Singh International Stadium	0

```
Sharjah Cricket Stadium                                          0
Sheikh Zayed Stadium                                             0
St George's Park                                                 0
Subrata Roy Sahara Stadium                                       0
SuperSport Park                                                  0
Vidarbha Cricket Association Stadium, Jamtha                     0
Wankhede Stadium                                                 0
Wankhede Stadium, Mumbai                                         4
Zayed Cricket Stadium, Abu Dhabi                                 0

winner                                                Kochi Tuskers Kerala  \
venue
Arun Jaitley Stadium                                                    0
Arun Jaitley Stadium, Delhi                                             0
Barabati Stadium                                                        0
Barsapara Cricket Stadium, Guwahati                                     0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                      0
Brabourne Stadium                                                       0
Brabourne Stadium, Mumbai                                               0
Buffalo Park                                                            0
De Beers Diamond Oval                                                   0
Dr DY Patil Sports Academy                                              0
Dr DY Patil Sports Academy, Mumbai                                      0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                      0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                      0
Dubai International Cricket Stadium                                     0
Eden Gardens                                                            1
Eden Gardens, Kolkata                                                   0
Feroz Shah Kotla                                                        1
Green Park                                                              0
Himachal Pradesh Cricket Association Stadium                            0
Himachal Pradesh Cricket Association Stadium, D…                      0
Holkar Cricket Stadium                                                  1
JSCA International Stadium Complex                                      0
Kingsmead                                                               0
M Chinnaswamy Stadium                                                   0
M Chinnaswamy Stadium, Bengaluru                                        0
M.Chinnaswamy Stadium                                                   0
MA Chidambaram Stadium                                                  0
MA Chidambaram Stadium, Chepauk                                         0
MA Chidambaram Stadium, Chepauk, Chennai                                0
Maharaja Yadavindra Singh International Cricket…                      0
Maharaja Yadavindra Singh International Cricket…                      0
Maharashtra Cricket Association Stadium                                 0
Maharashtra Cricket Association Stadium, Pune                           0
Narendra Modi Stadium, Ahmedabad                                        0
Nehru Stadium                                                           2
```

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New Wanderers Stadium                                                  0
Newlands                                                               0
OUTsurance Oval                                                        0
Punjab Cricket Association IS Bindra Stadium                           0
Punjab Cricket Association IS Bindra Stadium, M…                     0
Punjab Cricket Association IS Bindra Stadium, M…                     0
Punjab Cricket Association Stadium, Mohali                             0
Rajiv Gandhi International Stadium                                     0
Rajiv Gandhi International Stadium, Uppal                              0
Rajiv Gandhi International Stadium, Uppal, Hyde…                     0
Sardar Patel Stadium, Motera                                           0
Saurashtra Cricket Association Stadium                                 0
Sawai Mansingh Stadium                                                 0
Sawai Mansingh Stadium, Jaipur                                         0
Shaheed Veer Narayan Singh International Stadium                       0
Sharjah Cricket Stadium                                                0
Sheikh Zayed Stadium                                                   0
St George's Park                                                       0
Subrata Roy Sahara Stadium                                             0
SuperSport Park                                                        0
Vidarbha Cricket Association Stadium, Jamtha                           0
Wankhede Stadium                                                       1
Wankhede Stadium, Mumbai                                               0
Zayed Cricket Stadium, Abu Dhabi                                       0

winner                                          Kolkata Knight Riders  \
venue
Arun Jaitley Stadium                                                    0
Arun Jaitley Stadium, Delhi                                             1
Barabati Stadium                                                        3
Barsapara Cricket Stadium, Guwahati                                     1
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                      1
Brabourne Stadium                                                       0
Brabourne Stadium, Mumbai                                               0
Buffalo Park                                                            0
De Beers Diamond Oval                                                   0
Dr DY Patil Sports Academy                                              2
Dr DY Patil Sports Academy, Mumbai                                      1
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                      0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                      1
Dubai International Cricket Stadium                                     3
Eden Gardens                                                           45
Eden Gardens, Kolkata                                                   9
Feroz Shah Kotla                                                        5
Green Park                                                              0
Himachal Pradesh Cricket Association Stadium                            0
Himachal Pradesh Cricket Association Stadium, D…                      0
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Holkar Cricket Stadium                                                  1
JSCA International Stadium Complex                                      1
Kingsmead                                                               2
M Chinnaswamy Stadium                                                   6
M Chinnaswamy Stadium, Bengaluru                                        2
M.Chinnaswamy Stadium                                                   2
MA Chidambaram Stadium                                                  0
MA Chidambaram Stadium, Chepauk                                         2
MA Chidambaram Stadium, Chepauk, Chennai                                4
Maharaja Yadavindra Singh International Cricket…                      0
Maharaja Yadavindra Singh International Cricket…                      0
Maharashtra Cricket Association Stadium                                 3
Maharashtra Cricket Association Stadium, Pune                           2
Narendra Modi Stadium, Ahmedabad                                        3
Nehru Stadium                                                           0
New Wanderers Stadium                                                   0
Newlands                                                                0
OUTsurance Oval                                                         0
Punjab Cricket Association IS Bindra Stadium                            1
Punjab Cricket Association IS Bindra Stadium, M…                      1
Punjab Cricket Association IS Bindra Stadium, M…                      0
Punjab Cricket Association Stadium, Mohali                              2
Rajiv Gandhi International Stadium                                      1
Rajiv Gandhi International Stadium, Uppal                               4
Rajiv Gandhi International Stadium, Uppal, Hyde…                      1
Sardar Patel Stadium, Motera                                            0
Saurashtra Cricket Association Stadium                                  1
Sawai Mansingh Stadium                                                  3
Sawai Mansingh Stadium, Jaipur                                          0
Shaheed Veer Narayan Singh International Stadium                        0
Sharjah Cricket Stadium                                                 5
Sheikh Zayed Stadium                                                    5
St George's Park                                                        0
Subrata Roy Sahara Stadium                                              3
SuperSport Park                                                         1
Vidarbha Cricket Association Stadium, Jamtha                            0
Wankhede Stadium                                                        1
Wankhede Stadium, Mumbai                                                4
Zayed Cricket Stadium, Abu Dhabi                                        2

winner                                            Lucknow Super Giants  \
venue
Arun Jaitley Stadium                                                   0
Arun Jaitley Stadium, Delhi                                            0
Barabati Stadium                                                       0
Barsapara Cricket Stadium, Guwahati                                    0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                     9
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Brabourne Stadium	0
Brabourne Stadium, Mumbai	2
Buffalo Park	0
De Beers Diamond Oval	0
Dr DY Patil Sports Academy	0
Dr DY Patil Sports Academy, Mumbai	3
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dubai International Cricket Stadium	0
Eden Gardens	0
Eden Gardens, Kolkata	2
Feroz Shah Kotla	0
Green Park	0
Himachal Pradesh Cricket Association Stadium	0
Himachal Pradesh Cricket Association Stadium, D…	0
Holkar Cricket Stadium	0
JSCA International Stadium Complex	0
Kingsmead	0
M Chinnaswamy Stadium	0
M Chinnaswamy Stadium, Bengaluru	2
M.Chinnaswamy Stadium	0
MA Chidambaram Stadium	0
MA Chidambaram Stadium, Chepauk	0
MA Chidambaram Stadium, Chepauk, Chennai	1
Maharaja Yadavindra Singh International Cricket…	0
Maharaja Yadavindra Singh International Cricket…	0
Maharashtra Cricket Association Stadium	0
Maharashtra Cricket Association Stadium, Pune	2
Narendra Modi Stadium, Ahmedabad	1
Nehru Stadium	0
New Wanderers Stadium	0
Newlands	0
OUTsurance Oval	0
Punjab Cricket Association IS Bindra Stadium	0
Punjab Cricket Association IS Bindra Stadium, M…	0
Punjab Cricket Association IS Bindra Stadium, M…	1
Punjab Cricket Association Stadium, Mohali	0
Rajiv Gandhi International Stadium	0
Rajiv Gandhi International Stadium, Uppal	0
Rajiv Gandhi International Stadium, Uppal, Hyde…	2
Sardar Patel Stadium, Motera	0
Saurashtra Cricket Association Stadium	0
Sawai Mansingh Stadium	0
Sawai Mansingh Stadium, Jaipur	2
Shaheed Veer Narayan Singh International Stadium	0
Sharjah Cricket Stadium	0
Sheikh Zayed Stadium	0

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St George's Park                                                    0
Subrata Roy Sahara Stadium                                          0
SuperSport Park                                                     0
Vidarbha Cricket Association Stadium, Jamtha                        0
Wankhede Stadium                                                    0
Wankhede Stadium, Mumbai                                            3
Zayed Cricket Stadium, Abu Dhabi                                    0

winner                                              Mumbai Indians  \
venue
Arun Jaitley Stadium                                             1
Arun Jaitley Stadium, Delhi                                      4
Barabati Stadium                                                 0
Barsapara Cricket Stadium, Guwahati                              0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…               0
Brabourne Stadium                                                5
Brabourne Stadium, Mumbai                                        2
Buffalo Park                                                     1
De Beers Diamond Oval                                            0
Dr DY Patil Sports Academy                                       5
Dr DY Patil Sports Academy, Mumbai                               1
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…               2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…               0
Dubai International Cricket Stadium                              3
Eden Gardens                                                    10
Eden Gardens, Kolkata                                            0
Feroz Shah Kotla                                                 3
Green Park                                                       0
Himachal Pradesh Cricket Association Stadium                     0
Himachal Pradesh Cricket Association Stadium, D…               0
Holkar Cricket Stadium                                           2
JSCA International Stadium Complex                               0
Kingsmead                                                        0
M Chinnaswamy Stadium                                            8
M Chinnaswamy Stadium, Bengaluru                                 0
M.Chinnaswamy Stadium                                            1
MA Chidambaram Stadium                                           2
MA Chidambaram Stadium, Chepauk                                  3
MA Chidambaram Stadium, Chepauk, Chennai                         3
Maharaja Yadavindra Singh International Cricket…               1
Maharaja Yadavindra Singh International Cricket…               1
Maharashtra Cricket Association Stadium                          2
Maharashtra Cricket Association Stadium, Pune                    0
Narendra Modi Stadium, Ahmedabad                                 0
Nehru Stadium                                                    0
New Wanderers Stadium                                            0
Newlands                                                         1
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OUTsurance Oval                                                    0
Punjab Cricket Association IS Bindra Stadium                       0
Punjab Cricket Association IS Bindra Stadium, M…                 1
Punjab Cricket Association IS Bindra Stadium, M…                 1
Punjab Cricket Association Stadium, Mohali                         3
Rajiv Gandhi International Stadium                                 2
Rajiv Gandhi International Stadium, Uppal                          5
Rajiv Gandhi International Stadium, Uppal, Hyde…                 2
Sardar Patel Stadium, Motera                                       1
Saurashtra Cricket Association Stadium                             0
Sawai Mansingh Stadium                                             2
Sawai Mansingh Stadium, Jaipur                                     1
Shaheed Veer Narayan Singh International Stadium                   0
Sharjah Cricket Stadium                                            3
Sheikh Zayed Stadium                                               6
St George's Park                                                   2
Subrata Roy Sahara Stadium                                         2
SuperSport Park                                                    1
Vidarbha Cricket Association Stadium, Jamtha                       0
Wankhede Stadium                                                  41
Wankhede Stadium, Mumbai                                          15
Zayed Cricket Stadium, Abu Dhabi                                   2

winner                                                Pune Warriors  \
venue
Arun Jaitley Stadium                                              0
Arun Jaitley Stadium, Delhi                                       0
Barabati Stadium                                                  0
Barsapara Cricket Stadium, Guwahati                               0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                0
Brabourne Stadium                                                 0
Brabourne Stadium, Mumbai                                         0
Buffalo Park                                                      0
De Beers Diamond Oval                                             0
Dr DY Patil Sports Academy                                        2
Dr DY Patil Sports Academy, Mumbai                                0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                0
Dubai International Cricket Stadium                               0
Eden Gardens                                                      0
Eden Gardens, Kolkata                                             0
Feroz Shah Kotla                                                  1
Green Park                                                        0
Himachal Pradesh Cricket Association Stadium                      0
Himachal Pradesh Cricket Association Stadium, D…                0
Holkar Cricket Stadium                                            0
JSCA International Stadium Complex                                1
```

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Kingsmead                                                    0
M Chinnaswamy Stadium                                        0
M Chinnaswamy Stadium, Bengaluru                             0
M.Chinnaswamy Stadium                                        0
MA Chidambaram Stadium                                       0
MA Chidambaram Stadium, Chepauk                              1
MA Chidambaram Stadium, Chepauk, Chennai                     0
Maharaja Yadavindra Singh International Cricket…           0
Maharaja Yadavindra Singh International Cricket…           0
Maharashtra Cricket Association Stadium                      0
Maharashtra Cricket Association Stadium, Pune                0
Narendra Modi Stadium, Ahmedabad                             0
Nehru Stadium                                                0
New Wanderers Stadium                                        0
Newlands                                                     0
OUTsurance Oval                                              0
Punjab Cricket Association IS Bindra Stadium                 0
Punjab Cricket Association IS Bindra Stadium, M…           0
Punjab Cricket Association IS Bindra Stadium, M…           0
Punjab Cricket Association Stadium, Mohali                   1
Rajiv Gandhi International Stadium                           0
Rajiv Gandhi International Stadium, Uppal                    1
Rajiv Gandhi International Stadium, Uppal, Hyde…           0
Sardar Patel Stadium, Motera                                 0
Saurashtra Cricket Association Stadium                       0
Sawai Mansingh Stadium                                       0
Sawai Mansingh Stadium, Jaipur                               0
Shaheed Veer Narayan Singh International Stadium             0
Sharjah Cricket Stadium                                      0
Sheikh Zayed Stadium                                         0
St George's Park                                             0
Subrata Roy Sahara Stadium                                   4
SuperSport Park                                              0
Vidarbha Cricket Association Stadium, Jamtha                 0
Wankhede Stadium                                             1
Wankhede Stadium, Mumbai                                     0
Zayed Cricket Stadium, Abu Dhabi                             0

winner                                              Punjab Kings  \
venue
Arun Jaitley Stadium                                           1
Arun Jaitley Stadium, Delhi                                    1
Barabati Stadium                                               1
Barsapara Cricket Stadium, Guwahati                            2
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…             2
Brabourne Stadium                                              0
Brabourne Stadium, Mumbai                                      2
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Buffalo Park	0
De Beers Diamond Oval	1
Dr DY Patil Sports Academy	0
Dr DY Patil Sports Academy, Mumbai	2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	1
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dubai International Cricket Stadium	6
Eden Gardens	3
Eden Gardens, Kolkata	1
Feroz Shah Kotla	4
Green Park	0
Himachal Pradesh Cricket Association Stadium	5
Himachal Pradesh Cricket Association Stadium, D…	1
Holkar Cricket Stadium	4
JSCA International Stadium Complex	0
Kingsmead	2
M Chinnaswamy Stadium	5
M Chinnaswamy Stadium, Bengaluru	1
M.Chinnaswamy Stadium	0
MA Chidambaram Stadium	0
MA Chidambaram Stadium, Chepauk	1
MA Chidambaram Stadium, Chepauk, Chennai	4
Maharaja Yadavindra Singh International Cricket…	3
Maharaja Yadavindra Singh International Cricket…	0
Maharashtra Cricket Association Stadium	0
Maharashtra Cricket Association Stadium, Pune	1
Narendra Modi Stadium, Ahmedabad	4
Nehru Stadium	0
New Wanderers Stadium	1
Newlands	1
OUTsurance Oval	1
Punjab Cricket Association IS Bindra Stadium	8
Punjab Cricket Association IS Bindra Stadium, M…	4
Punjab Cricket Association IS Bindra Stadium, M…	1
Punjab Cricket Association Stadium, Mohali	18
Rajiv Gandhi International Stadium	0
Rajiv Gandhi International Stadium, Uppal	4
Rajiv Gandhi International Stadium, Uppal, Hyde…	0
Sardar Patel Stadium, Motera	0
Saurashtra Cricket Association Stadium	2
Sawai Mansingh Stadium	1
Sawai Mansingh Stadium, Jaipur	2
Shaheed Veer Narayan Singh International Stadium	0
Sharjah Cricket Stadium	5
Sheikh Zayed Stadium	2
St George's Park	1
Subrata Roy Sahara Stadium	1

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SuperSport Park                                                  0
Vidarbha Cricket Association Stadium, Jamtha                     0
Wankhede Stadium                                                 5
Wankhede Stadium, Mumbai                                         4
Zayed Cricket Stadium, Abu Dhabi                                 0

winner                                                Rajasthan Royals  \
venue
Arun Jaitley Stadium                                                0
Arun Jaitley Stadium, Delhi                                         2
Barabati Stadium                                                    0
Barsapara Cricket Stadium, Guwahati                                 2
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                  1
Brabourne Stadium                                                   2
Brabourne Stadium, Mumbai                                           3
Buffalo Park                                                        0
De Beers Diamond Oval                                               0
Dr DY Patil Sports Academy                                          1
Dr DY Patil Sports Academy, Mumbai                                  1
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                  1
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                  0
Dubai International Cricket Stadium                                 2
Eden Gardens                                                        2
Eden Gardens, Kolkata                                               2
Feroz Shah Kotla                                                    4
Green Park                                                          0
Himachal Pradesh Cricket Association Stadium                        0
Himachal Pradesh Cricket Association Stadium, D…                  1
Holkar Cricket Stadium                                              0
JSCA International Stadium Complex                                  0
Kingsmead                                                           2
M Chinnaswamy Stadium                                               3
M Chinnaswamy Stadium, Bengaluru                                    0
M.Chinnaswamy Stadium                                               1
MA Chidambaram Stadium                                              0
MA Chidambaram Stadium, Chepauk                                     1
MA Chidambaram Stadium, Chepauk, Chennai                            1
Maharaja Yadavindra Singh International Cricket…                  2
Maharaja Yadavindra Singh International Cricket…                  0
Maharashtra Cricket Association Stadium                             1
Maharashtra Cricket Association Stadium, Pune                       2
Narendra Modi Stadium, Ahmedabad                                    3
Nehru Stadium                                                       0
New Wanderers Stadium                                               0
Newlands                                                            0
OUTsurance Oval                                                     0
Punjab Cricket Association IS Bindra Stadium                        0
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Punjab Cricket Association IS Bindra Stadium, M…                 0
Punjab Cricket Association IS Bindra Stadium, M…                 0
Punjab Cricket Association Stadium, Mohali                         3
Rajiv Gandhi International Stadium                                 0
Rajiv Gandhi International Stadium, Uppal                          2
Rajiv Gandhi International Stadium, Uppal, Hyde…                 1
Sardar Patel Stadium, Motera                                       7
Saurashtra Cricket Association Stadium                             0
Sawai Mansingh Stadium                                            32
Sawai Mansingh Stadium, Jaipur                                     6
Shaheed Veer Narayan Singh International Stadium                   0
Sharjah Cricket Stadium                                            2
Sheikh Zayed Stadium                                               5
St George's Park                                                   1
Subrata Roy Sahara Stadium                                         1
SuperSport Park                                                    2
Vidarbha Cricket Association Stadium, Jamtha                       1
Wankhede Stadium                                                   4
Wankhede Stadium, Mumbai                                           6
Zayed Cricket Stadium, Abu Dhabi                                   1

winner                                              Rising Pune Supergiant  \
venue
Arun Jaitley Stadium                                                   0
Arun Jaitley Stadium, Delhi                                            0
Barabati Stadium                                                       0
Barsapara Cricket Stadium, Guwahati                                    0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                     0
Brabourne Stadium                                                      0
Brabourne Stadium, Mumbai                                              0
Buffalo Park                                                           0
De Beers Diamond Oval                                                  0
Dr DY Patil Sports Academy                                             0
Dr DY Patil Sports Academy, Mumbai                                     0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                     2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                     0
Dubai International Cricket Stadium                                    0
Eden Gardens                                                           1
Eden Gardens, Kolkata                                                  0
Feroz Shah Kotla                                                       1
Green Park                                                             0
Himachal Pradesh Cricket Association Stadium                           0
Himachal Pradesh Cricket Association Stadium, D…                     0
Holkar Cricket Stadium                                                 0
JSCA International Stadium Complex                                     0
Kingsmead                                                              0
M Chinnaswamy Stadium                                                  1
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M Chinnaswamy Stadium, Bengaluru                                  0
M.Chinnaswamy Stadium                                             0
MA Chidambaram Stadium                                            0
MA Chidambaram Stadium, Chepauk                                   0
MA Chidambaram Stadium, Chepauk, Chennai                          0
Maharaja Yadavindra Singh International Cricket…                0
Maharaja Yadavindra Singh International Cricket…                0
Maharashtra Cricket Association Stadium                           5
Maharashtra Cricket Association Stadium, Pune                     0
Narendra Modi Stadium, Ahmedabad                                  0
Nehru Stadium                                                     0
New Wanderers Stadium                                             0
Newlands                                                          0
OUTsurance Oval                                                   0
Punjab Cricket Association IS Bindra Stadium                      0
Punjab Cricket Association IS Bindra Stadium, M…                0
Punjab Cricket Association IS Bindra Stadium, M…                0
Punjab Cricket Association Stadium, Mohali                        0
Rajiv Gandhi International Stadium                                0
Rajiv Gandhi International Stadium, Uppal                         2
Rajiv Gandhi International Stadium, Uppal, Hyde…                0
Sardar Patel Stadium, Motera                                      0
Saurashtra Cricket Association Stadium                            0
Sawai Mansingh Stadium                                            0
Sawai Mansingh Stadium, Jaipur                                    0
Shaheed Veer Narayan Singh International Stadium                  0
Sharjah Cricket Stadium                                           0
Sheikh Zayed Stadium                                              0
St George's Park                                                  0
Subrata Roy Sahara Stadium                                        0
SuperSport Park                                                   0
Vidarbha Cricket Association Stadium, Jamtha                      0
Wankhede Stadium                                                  3
Wankhede Stadium, Mumbai                                          0
Zayed Cricket Stadium, Abu Dhabi                                  0

winner                                          Royal Challengers Bengaluru
\
venue
Arun Jaitley Stadium                                                    1
Arun Jaitley Stadium, Delhi                                             1
Barabati Stadium                                                        0
Barsapara Cricket Stadium, Guwahati                                     0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                      2
Brabourne Stadium                                                       1
Brabourne Stadium, Mumbai                                               0
Buffalo Park                                                            0
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De Beers Diamond Oval	0
Dr DY Patil Sports Academy	1
Dr DY Patil Sports Academy, Mumbai	2
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…	0
Dubai International Cricket Stadium	7
Eden Gardens	4
Eden Gardens, Kolkata	2
Feroz Shah Kotla	5
Green Park	0
Himachal Pradesh Cricket Association Stadium	0
Himachal Pradesh Cricket Association Stadium, D…	1
Holkar Cricket Stadium	1
JSCA International Stadium Complex	1
Kingsmead	3
M Chinnaswamy Stadium	28
M Chinnaswamy Stadium, Bengaluru	9
M.Chinnaswamy Stadium	8
MA Chidambaram Stadium	0
MA Chidambaram Stadium, Chepauk	2
MA Chidambaram Stadium, Chepauk, Chennai	4
Maharaja Yadavindra Singh International Cricket…	2
Maharaja Yadavindra Singh International Cricket…	0
Maharashtra Cricket Association Stadium	2
Maharashtra Cricket Association Stadium, Pune	2
Narendra Modi Stadium, Ahmedabad	3
Nehru Stadium	1
New Wanderers Stadium	3
Newlands	1
OUTsurance Oval	0
Punjab Cricket Association IS Bindra Stadium	1
Punjab Cricket Association IS Bindra Stadium, M…	1
Punjab Cricket Association IS Bindra Stadium, M…	1
Punjab Cricket Association Stadium, Mohali	2
Rajiv Gandhi International Stadium	0
Rajiv Gandhi International Stadium, Uppal	2
Rajiv Gandhi International Stadium, Uppal, Hyde…	2
Sardar Patel Stadium, Motera	1
Saurashtra Cricket Association Stadium	1
Sawai Mansingh Stadium	3
Sawai Mansingh Stadium, Jaipur	2
Shaheed Veer Narayan Singh International Stadium	1
Sharjah Cricket Stadium	3
Sheikh Zayed Stadium	2
St George's Park	0
Subrata Roy Sahara Stadium	2
SuperSport Park	2

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Vidarbha Cricket Association Stadium, Jamtha                                0
Wankhede Stadium                                                            3
Wankhede Stadium, Mumbai                                                    6
Zayed Cricket Stadium, Abu Dhabi                                            0

winner                                                Sunrisers Hyderabad
venue
Arun Jaitley Stadium                                                      2
Arun Jaitley Stadium, Delhi                                               3
Barabati Stadium                                                          0
Barsapara Cricket Stadium, Guwahati                                       0
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cr…                    2
Brabourne Stadium                                                         1
Brabourne Stadium, Mumbai                                                 2
Buffalo Park                                                              0
De Beers Diamond Oval                                                     0
Dr DY Patil Sports Academy                                                0
Dr DY Patil Sports Academy, Mumbai                                        3
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                    3
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Sta…                    0
Dubai International Cricket Stadium                                       7
Eden Gardens                                                              2
Eden Gardens, Kolkata                                                     1
Feroz Shah Kotla                                                          4
Green Park                                                                1
Himachal Pradesh Cricket Association Stadium                              0
Himachal Pradesh Cricket Association Stadium, D…                    0
Holkar Cricket Stadium                                                    0
JSCA International Stadium Complex                                        1
Kingsmead                                                                 0
M Chinnaswamy Stadium                                                     2
M Chinnaswamy Stadium, Bengaluru                                          1
M.Chinnaswamy Stadium                                                     0
MA Chidambaram Stadium                                                    0
MA Chidambaram Stadium, Chepauk                                           0
MA Chidambaram Stadium, Chepauk, Chennai                                  3
Maharaja Yadavindra Singh International Cricket…                    1
Maharaja Yadavindra Singh International Cricket…                    0
Maharashtra Cricket Association Stadium                                   1
Maharashtra Cricket Association Stadium, Pune                             0
Narendra Modi Stadium, Ahmedabad                                          0
Nehru Stadium                                                             0
New Wanderers Stadium                                                     0
Newlands                                                                  0
OUTsurance Oval                                                           0
Punjab Cricket Association IS Bindra Stadium                              0
Punjab Cricket Association IS Bindra Stadium, M…                    2
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Punjab Cricket Association IS Bindra Stadium, M…                    0
Punjab Cricket Association Stadium, Mohali                            2
Rajiv Gandhi International Stadium                                   10
Rajiv Gandhi International Stadium, Uppal                            19
Rajiv Gandhi International Stadium, Uppal, Hyde…                    8
Sardar Patel Stadium, Motera                                          1
Saurashtra Cricket Association Stadium                                1
Sawai Mansingh Stadium                                                1
Sawai Mansingh Stadium, Jaipur                                        1
Shaheed Veer Narayan Singh International Stadium                      1
Sharjah Cricket Stadium                                               2
Sheikh Zayed Stadium                                                  2
St George's Park                                                      0
Subrata Roy Sahara Stadium                                            0
SuperSport Park                                                       0
Vidarbha Cricket Association Stadium, Jamtha                          0
Wankhede Stadium                                                      1
Wankhede Stadium, Mumbai                                              1
Zayed Cricket Stadium, Abu Dhabi                                      1
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[16]:
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data_df['venue'].value_counts()
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[16]:
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venue
Eden Gardens                                                          77
Wankhede Stadium                                                      72
M Chinnaswamy Stadium                                                 62
Feroz Shah Kotla                                                      59
Wankhede Stadium, Mumbai                                              52
Rajiv Gandhi International Stadium, Uppal                             48
Sawai Mansingh Stadium                                                47
MA Chidambaram Stadium, Chepauk                                       47
Dubai International Cricket Stadium                                   43
Punjab Cricket Association Stadium, Mohali                            35
MA Chidambaram Stadium, Chepauk, Chennai                              33
Narendra Modi Stadium, Ahmedabad                                      33
Sharjah Cricket Stadium                                               28
Sheikh Zayed Stadium                                                  27
Maharashtra Cricket Association Stadium                               22
Arun Jaitley Stadium, Delhi                                           22
Eden Gardens, Kolkata                                                 22
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium, Lucknow 21
Dr DY Patil Sports Academy, Mumbai                                    20
M Chinnaswamy Stadium, Bengaluru                                      19
Rajiv Gandhi International Stadium, Uppal, Hyderabad                  18
Dr DY Patil Sports Academy                                            17
Brabourne Stadium, Mumbai                                             17
Sawai Mansingh Stadium, Jaipur                                        17
```
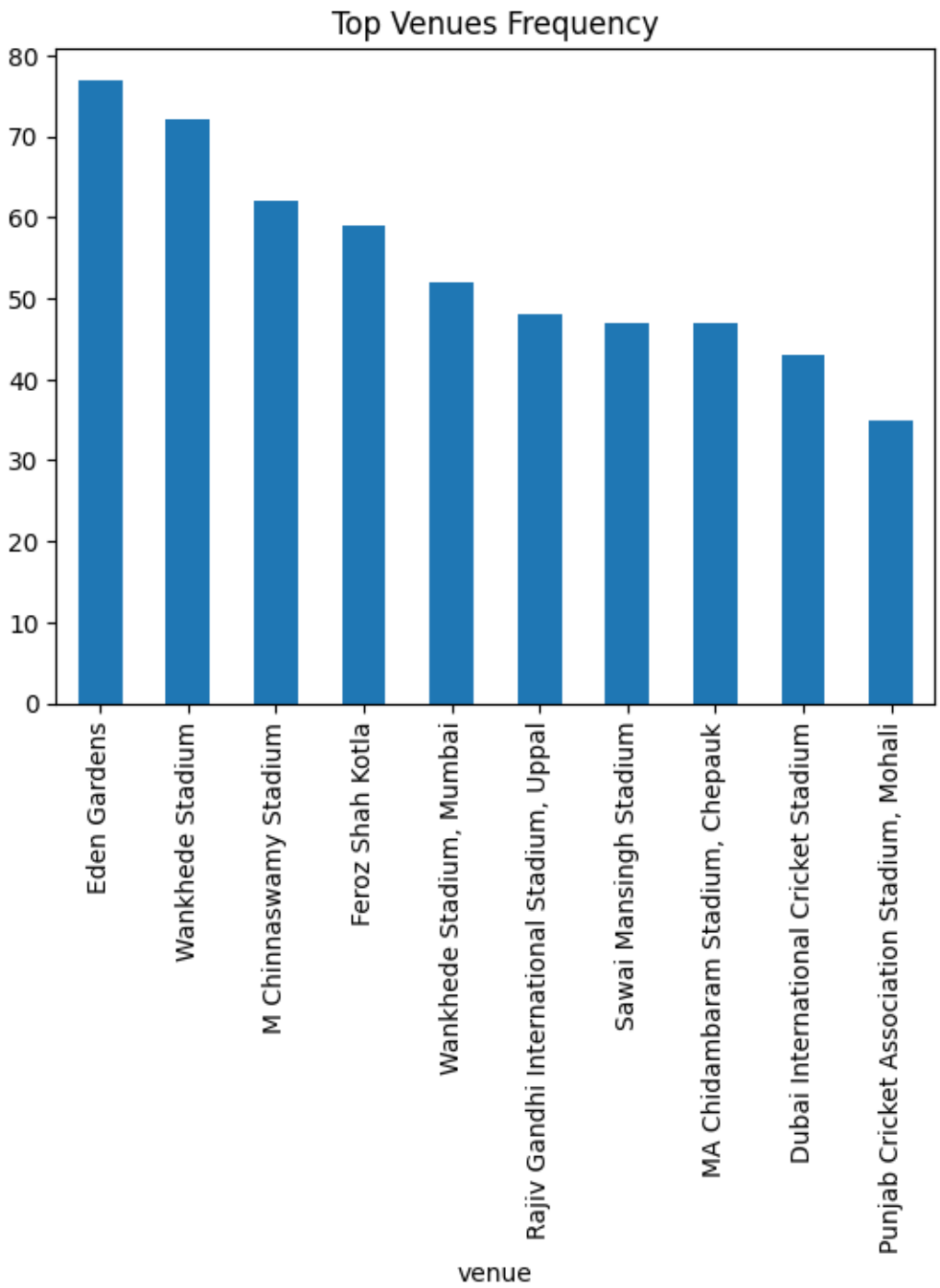
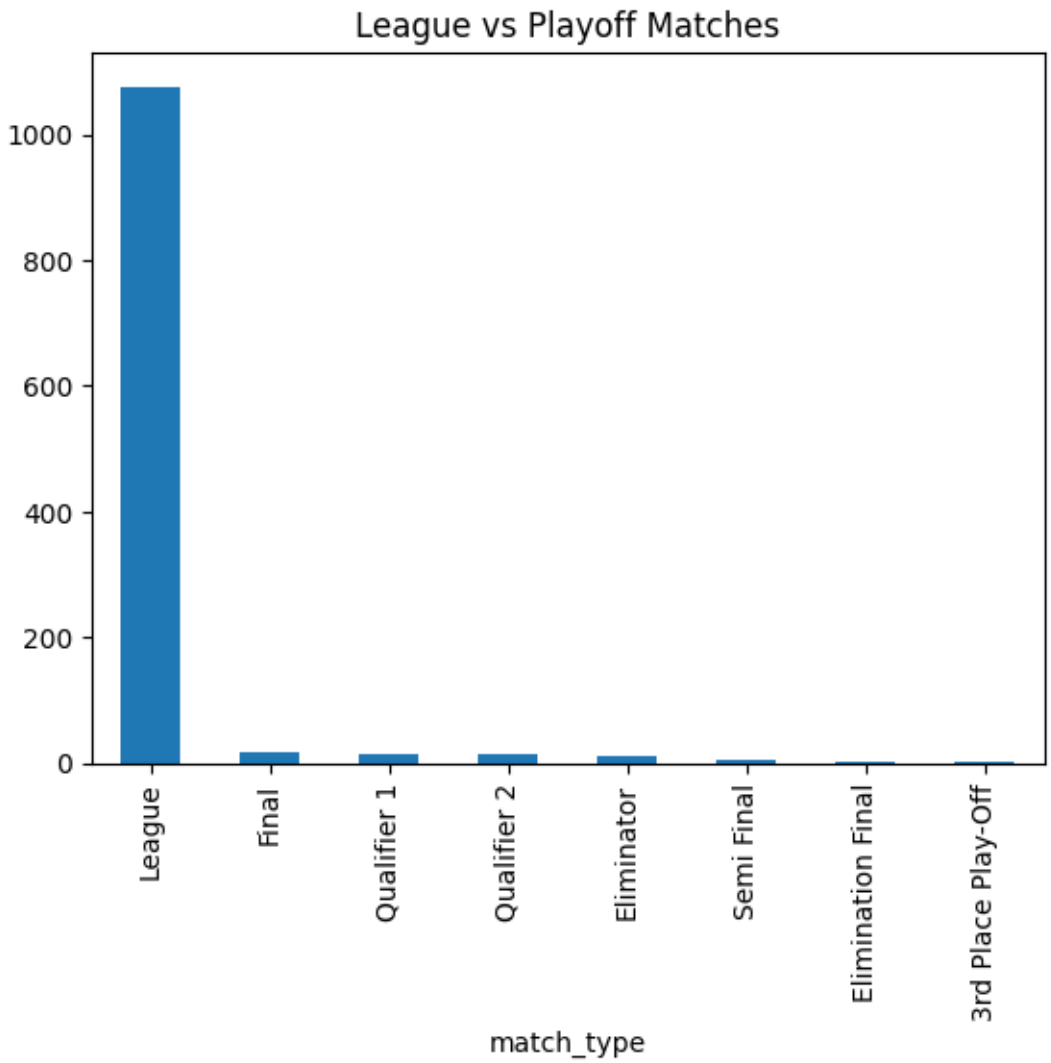

```
Subrata Roy Sahara Stadium                                          16
Kingsmead                                                           15
Rajiv Gandhi International Stadium                                  15
M.Chinnaswamy Stadium                                               14
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium                 13
Arun Jaitley Stadium                                                13
Maharashtra Cricket Association Stadium, Pune                       13
SuperSport Park                                                     12
Sardar Patel Stadium, Motera                                        11
Punjab Cricket Association IS Bindra Stadium, Mohali                11
Brabourne Stadium                                                   10
Punjab Cricket Association IS Bindra Stadium                        10
Maharaja Yadavindra Singh International Cricket Stadium, Mullanpur  10
Himachal Pradesh Cricket Association Stadium                         9
Holkar Cricket Stadium                                               9
Saurashtra Cricket Association Stadium                               9
MA Chidambaram Stadium                                               9
New Wanderers Stadium                                                8
Zayed Cricket Stadium, Abu Dhabi                                     8
St George's Park                                                     7
Barabati Stadium                                                     7
JSCA International Stadium Complex                                   7
Newlands                                                             6
Shaheed Veer Narayan Singh International Stadium                     6
Nehru Stadium                                                        5
Punjab Cricket Association IS Bindra Stadium, Mohali, Chandigarh     5
Barsapara Cricket Stadium, Guwahati                                  5
Himachal Pradesh Cricket Association Stadium, Dharamsala             5
Green Park                                                           4
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium, Visakhapatnam   4
Buffalo Park                                                         3
De Beers Diamond Oval                                                3
Vidarbha Cricket Association Stadium, Jamtha                         3
OUTsurance Oval                                                      2
Maharaja Yadavindra Singh International Cricket Stadium, New Chandigarh  1
Name: count, dtype: int64
```

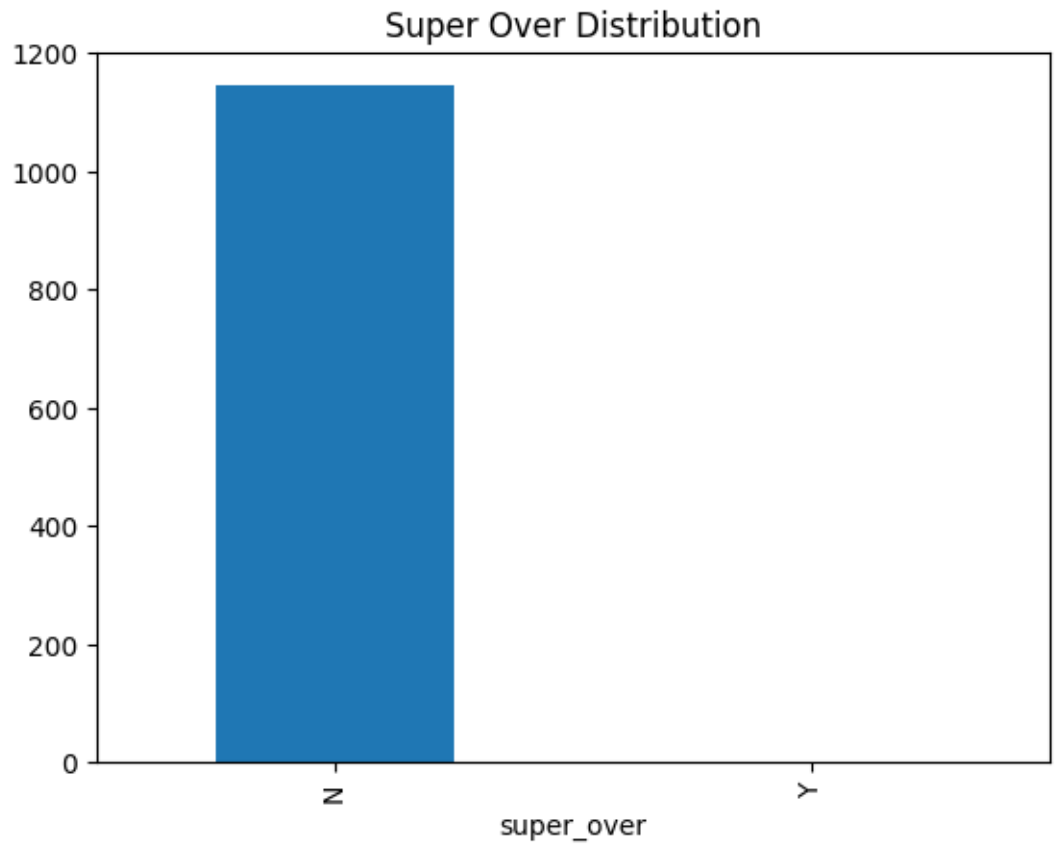
[17]:
```
data_df['venue'].value_counts().head(10).plot(kind='bar')
plt.title("Top Venues Frequency")
plt.show()
```



[18]:
```
data_df['match_type'].value_counts().plot(kind='bar')
plt.title("League vs Playoff Matches")
plt.show()
```



[19]:
```
data_df['super_over'].value_counts().plot(kind='bar')
plt.title("Super Over Distribution")
plt.show()
```



[20]:
```
obsolete_teams = [
    'Deccan Chargers',
    'Gujarat Lions',
    'Pune Warriors',
    'Kochi Tuskers Kerala',
    'Rising Pune Supergiant'
]

for col in ['team1', 'team2', 'toss_winner', 'winner']:
    data_df = data_df[~data_df[col].isin(obsolete_teams)]
```

[21]:
```
data_df.drop([
    'player_of_match',
    'result',
    'result_margin',
    'match_date',
    'team1_players',
    'team2_players'
], axis=1, inplace=True)
```

[22]:
```
data_df.head(10)
```

[22]:
```
                          team1                        team2  \
0   Royal Challengers Bengaluru        Kolkata Knight Riders
1                  Punjab Kings          Chennai Super Kings
2                Delhi Capitals             Rajasthan Royals
3                Mumbai Indians  Royal Challengers Bengaluru
5               Rajasthan Royals                 Punjab Kings
7            Chennai Super Kings               Mumbai Indians
9                  Punjab Kings               Mumbai Indians
10           Chennai Super Kings        Kolkata Knight Riders
11  Royal Challengers Bengaluru             Rajasthan Royals
13                 Punjab Kings               Delhi Capitals

                    toss_winner toss_decision                       winner  \
0   Royal Challengers Bengaluru         field        Kolkata Knight Riders
1            Chennai Super Kings           bat          Chennai Super Kings
2               Rajasthan Royals           bat               Delhi Capitals
3                 Mumbai Indians           bat  Royal Challengers Bengaluru
5                  Punjab Kings           bat             Rajasthan Royals
7                 Mumbai Indians         field          Chennai Super Kings
9                 Mumbai Indians         field                 Punjab Kings
10        Kolkata Knight Riders           bat          Chennai Super Kings
11              Rajasthan Royals         field             Rajasthan Royals
13                Delhi Capitals           bat                 Punjab Kings

                                         venue        city  season  \
0                       M Chinnaswamy Stadium   Bangalore       1
1   Punjab Cricket Association Stadium, Mohali  Chandigarh       1
2                             Feroz Shah Kotla       Delhi       1
3                             Wankhede Stadium      Mumbai       1
5                       Sawai Mansingh Stadium      Jaipur       1
7               MA Chidambaram Stadium, Chepauk     Chennai       1
9   Punjab Cricket Association Stadium, Mohali  Chandigarh       1
10              MA Chidambaram Stadium, Chepauk     Chennai       1
11                      M Chinnaswamy Stadium   Bangalore       1
13  Punjab Cricket Association Stadium, Mohali  Chandigarh       1

    match_number match_type  target_runs  target_overs super_over
0              1     League        223.0          20.0          N
1              2     League        241.0          20.0          N
2              3     League        130.0          20.0          N
3              4     League        166.0          20.0          N
5              6     League        167.0          20.0          N
7              8     League        209.0          20.0          N
9             10     League        183.0          20.0          N
10            11     League        136.0          20.0          N
```

```
11            12     League        148.0          20.0          N
13            14     League        159.0          20.0          N
```

[23]:
```python
data_df['target_runs'] = data_df['target_runs'].fillna(0)
data_df['target_overs'] = data_df['target_overs'].fillna(0)
```

[24]:
```python
data_df.drop('city', axis=1, inplace=True)
```

[25]:
```python
data_df.head(4)
```

[25]:
```
                        team1                        team2  \
0  Royal Challengers Bengaluru        Kolkata Knight Riders
1                 Punjab Kings          Chennai Super Kings
2               Delhi Capitals             Rajasthan Royals
3               Mumbai Indians  Royal Challengers Bengaluru

                   toss_winner toss_decision                       winner  \
0  Royal Challengers Bengaluru         field        Kolkata Knight Riders
1          Chennai Super Kings           bat          Chennai Super Kings
2             Rajasthan Royals           bat                Delhi Capitals
3               Mumbai Indians           bat  Royal Challengers Bengaluru

                                       venue  season  match_number  \
0                       M Chinnaswamy Stadium       1             1
1  Punjab Cricket Association Stadium, Mohali       1             2
2                            Feroz Shah Kotla       1             3
3                            Wankhede Stadium       1             4

  match_type  target_runs  target_overs super_over
0     League        223.0          20.0          N
1     League        241.0          20.0          N
2     League        130.0          20.0          N
3     League        166.0          20.0          N
```

[26]:
```python
data_df.drop('match_number', axis=1, inplace=True)
```

[27]:
```python
data_df['target_runs'] = data_df['target_runs'].astype(int)
data_df['target_overs'] = data_df['target_overs'].astype(int)
```

[28]:
```python
features = data_df.drop('winner', axis=1)
target = data_df['winner']
```

[29]:
```python
categorical_cols = [
    'team1',
    'team2',
    'toss_winner',
    'toss_decision',
```

```
    'venue',
    'match_type',
    'super_over'
]

numerical_cols = [
    'season',
    'target_runs',
    'target_overs'
]
```

[30]:
```
x_train, x_test, y_train, y_test = train_test_split(features, target,␣
↪test_size=0.2, random_state=365, stratify=target)
```

[31]:
```
categorical_coloumns = Pipeline(steps=[('onehot',␣
↪OneHotEncoder(handle_unknown='ignore'))])
numerical_coloumns = Pipeline(steps=[('imputer',␣
↪SimpleImputer(strategy='median'))])
```

[32]:
```
preprocessor = ColumnTransformer(transformers=[('cat', categorical_coloumns,␣
↪categorical_cols),
                                               ('num', numerical_coloumns,␣
↪numerical_cols)])
```

[33]:
```
rf_pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                              ('model',␣
↪RandomForestClassifier(class_weight='balanced', random_state=365))])
```

[34]:
```
parameter = {'model__n_estimators': [200, 400, 600],
    'model__max_depth': [10, 20, 30, None],
    'model__min_samples_split': [2, 5, 10],
    'model__min_samples_leaf': [1, 2, 4],
    'model__max_features': ['sqrt', 'log2'],
    'model__bootstrap': [True, False]}
```

[35]:
```
gridsearch = GridSearchCV(rf_pipeline, param_grid=parameter,␣
↪cv=5,scoring='f1_weighted', n_jobs=-1)
```

[36]:
```
gridsearch.fit(x_train, y_train)
```

[36]:
```
GridSearchCV(cv=5,
             estimator=Pipeline(steps=[('preprocessor',
                                        ColumnTransformer(transformers=[('cat',
Pipeline(steps=[('onehot',
          OneHotEncoder(handle_unknown='ignore'))]),
['team1',
'team2',
```

```
'toss_winner',
'toss_decision',
'venue',
'match_type',
'super_over']),
                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
('num',
Pipeline(steps=[('imputer',
          SimpleImputer(strategy='median'))]),
['season',
'target_runs',
'target_overs'])])),
                                        ('model',
RandomForestClassifier(class_weight='balanced',
random_state=365))]),
             n_jobs=-1,
             param_grid={'model__bootstrap': [True, False],
                         'model__max_depth': [10, 20, 30, None],
                         'model__max_features': ['sqrt', 'log2'],
                         'model__min_samples_leaf': [1, 2, 4],
                         'model__min_samples_split': [2, 5, 10],
                         'model__n_estimators': [200, 400, 600]},
             scoring='f1_weighted')
```

[37]:
```
best_model = gridsearch.best_estimator_
```

[38]:
```
y_predicted = best_model.predict(x_test)
```

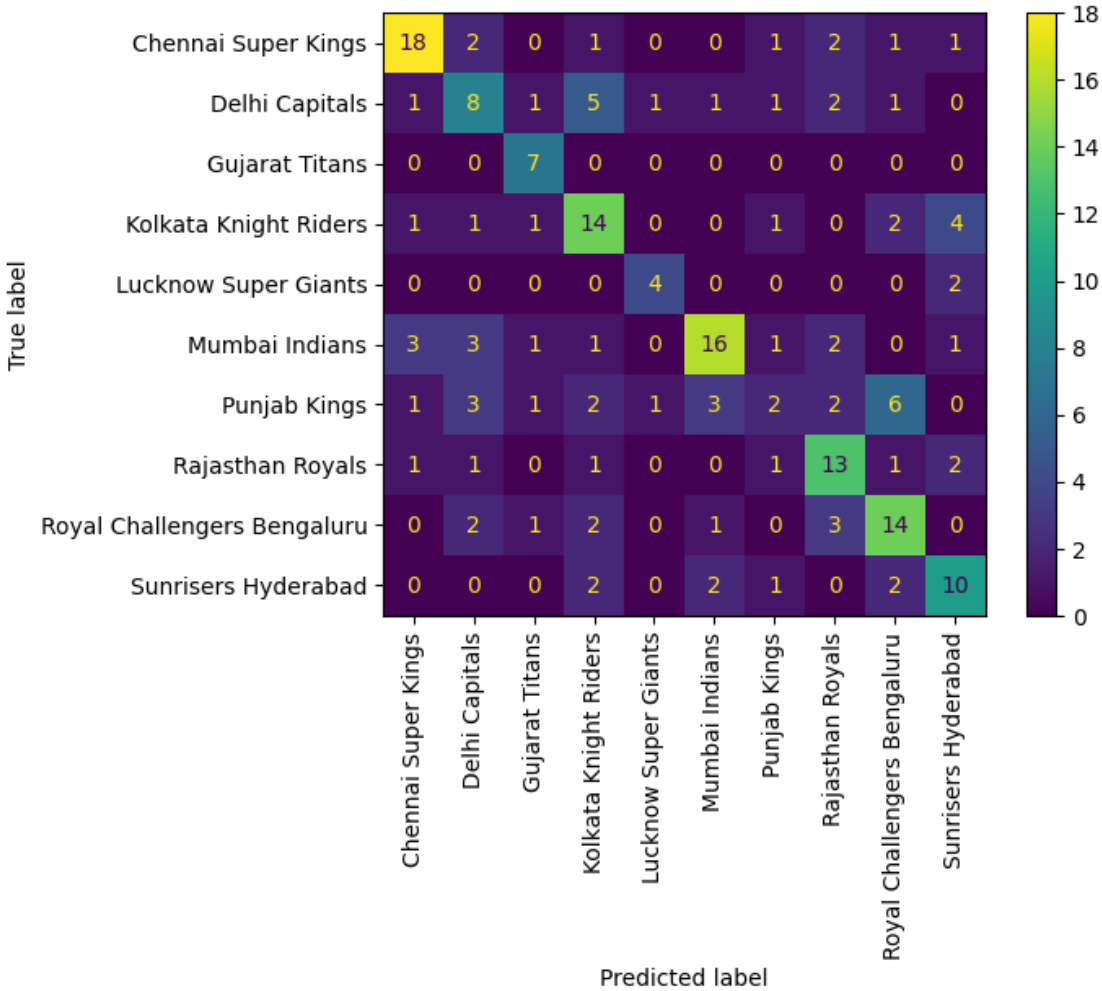
[39]:
```
y_predicted
```

[39]:
```
array(['Chennai Super Kings', 'Royal Challengers Bengaluru',
       'Kolkata Knight Riders', 'Rajasthan Royals', 'Sunrisers Hyderabad',
       'Delhi Capitals', 'Chennai Super Kings', 'Chennai Super Kings',
       'Sunrisers Hyderabad', 'Gujarat Titans', 'Lucknow Super Giants',
       'Kolkata Knight Riders', 'Mumbai Indians', 'Rajasthan Royals',
       'Chennai Super Kings', 'Mumbai Indians', 'Chennai Super Kings',
       'Royal Challengers Bengaluru', 'Delhi Capitals',
       'Chennai Super Kings', 'Chennai Super Kings', 'Punjab Kings',
       'Royal Challengers Bengaluru', 'Rajasthan Royals',
       'Rajasthan Royals', 'Lucknow Super Giants', 'Chennai Super Kings',
       'Mumbai Indians', 'Mumbai Indians', 'Punjab Kings',
       'Lucknow Super Giants', 'Punjab Kings', 'Sunrisers Hyderabad',
       'Sunrisers Hyderabad', 'Gujarat Titans', 'Rajasthan Royals',
       'Kolkata Knight Riders', 'Rajasthan Royals',
       'Kolkata Knight Riders', 'Royal Challengers Bengaluru',
       'Chennai Super Kings', 'Chennai Super Kings',
       'Kolkata Knight Riders', 'Chennai Super Kings', 'Mumbai Indians',
       'Mumbai Indians', 'Kolkata Knight Riders', 'Gujarat Titans',
```

```
'Kolkata Knight Riders', 'Royal Challengers Bengaluru',
'Lucknow Super Giants', 'Punjab Kings', 'Mumbai Indians',
'Sunrisers Hyderabad', 'Kolkata Knight Riders', 'Delhi Capitals',
'Mumbai Indians', 'Gujarat Titans', 'Sunrisers Hyderabad',
'Mumbai Indians', 'Royal Challengers Bengaluru',
'Royal Challengers Bengaluru', 'Delhi Capitals',
'Kolkata Knight Riders', 'Royal Challengers Bengaluru',
'Sunrisers Hyderabad', 'Royal Challengers Bengaluru',
'Delhi Capitals', 'Rajasthan Royals',
'Royal Challengers Bengaluru', 'Kolkata Knight Riders',
'Kolkata Knight Riders', 'Delhi Capitals', 'Chennai Super Kings',
'Rajasthan Royals', 'Rajasthan Royals', 'Chennai Super Kings',
'Royal Challengers Bengaluru', 'Mumbai Indians', 'Delhi Capitals',
'Kolkata Knight Riders', 'Sunrisers Hyderabad', 'Delhi Capitals',
'Gujarat Titans', 'Chennai Super Kings', 'Rajasthan Royals',
'Rajasthan Royals', 'Delhi Capitals',
'Royal Challengers Bengaluru', 'Mumbai Indians',
'Sunrisers Hyderabad', 'Chennai Super Kings', 'Gujarat Titans',
'Kolkata Knight Riders', 'Kolkata Knight Riders',
'Sunrisers Hyderabad', 'Royal Challengers Bengaluru',
'Sunrisers Hyderabad', 'Punjab Kings', 'Chennai Super Kings',
'Mumbai Indians', 'Rajasthan Royals', 'Rajasthan Royals',
'Delhi Capitals', 'Kolkata Knight Riders', 'Delhi Capitals',
'Royal Challengers Bengaluru', 'Delhi Capitals', 'Delhi Capitals',
'Delhi Capitals', 'Chennai Super Kings', 'Mumbai Indians',
'Royal Challengers Bengaluru', 'Rajasthan Royals', 'Punjab Kings',
'Mumbai Indians', 'Mumbai Indians', 'Kolkata Knight Riders',
'Rajasthan Royals', 'Royal Challengers Bengaluru',
'Royal Challengers Bengaluru', 'Rajasthan Royals',
'Sunrisers Hyderabad', 'Royal Challengers Bengaluru',
'Sunrisers Hyderabad', 'Mumbai Indians', 'Delhi Capitals',
'Chennai Super Kings', 'Rajasthan Royals', 'Kolkata Knight Riders',
'Sunrisers Hyderabad', 'Royal Challengers Bengaluru',
'Kolkata Knight Riders', 'Delhi Capitals', 'Sunrisers Hyderabad',
'Chennai Super Kings', 'Delhi Capitals', 'Gujarat Titans',
'Gujarat Titans', 'Punjab Kings', 'Royal Challengers Bengaluru',
'Kolkata Knight Riders', 'Sunrisers Hyderabad',
'Chennai Super Kings', 'Mumbai Indians', 'Kolkata Knight Riders',
'Royal Challengers Bengaluru', 'Chennai Super Kings',
'Sunrisers Hyderabad', 'Delhi Capitals', 'Kolkata Knight Riders',
'Chennai Super Kings', 'Chennai Super Kings',
'Royal Challengers Bengaluru', 'Rajasthan Royals',
'Rajasthan Royals', 'Rajasthan Royals', 'Rajasthan Royals',
'Punjab Kings', 'Lucknow Super Giants', 'Mumbai Indians',
'Mumbai Indians', 'Royal Challengers Bengaluru', 'Delhi Capitals',
'Gujarat Titans', 'Kolkata Knight Riders',
'Royal Challengers Bengaluru', 'Kolkata Knight Riders',
```

```
'Kolkata Knight Riders', 'Royal Challengers Bengaluru',
'Sunrisers Hyderabad', 'Rajasthan Royals', 'Sunrisers Hyderabad',
'Sunrisers Hyderabad', 'Rajasthan Royals', 'Kolkata Knight Riders',
'Mumbai Indians', 'Delhi Capitals', 'Kolkata Knight Riders',
'Rajasthan Royals', 'Gujarat Titans', 'Mumbai Indians',
'Gujarat Titans', 'Chennai Super Kings', 'Mumbai Indians',
'Chennai Super Kings', 'Royal Challengers Bengaluru',
'Mumbai Indians', 'Kolkata Knight Riders', 'Gujarat Titans',
'Royal Challengers Bengaluru', 'Lucknow Super Giants',
'Kolkata Knight Riders'], dtype=object)
```

[40]:
```
plt.Figure(figsize=(16,12))
ConfusionMatrixDisplay.from_predictions(y_test, y_predicted)
plt.xticks(rotation=90)
plt.show()
```



[41]:
```
print(classification_report(y_test, y_predicted))
```

```
                              precision    recall  f1-score   support

        Chennai Super Kings       0.72      0.69      0.71        26
             Delhi Capitals       0.40      0.38      0.39        21
             Gujarat Titans       0.58      1.00      0.74         7
      Kolkata Knight Riders       0.50      0.58      0.54        24
       Lucknow Super Giants       0.67      0.67      0.67         6
             Mumbai Indians       0.70      0.57      0.63        28
               Punjab Kings       0.25      0.10      0.14        21
           Rajasthan Royals       0.54      0.65      0.59        20
Royal Challengers Bengaluru       0.52      0.61      0.56        23
        Sunrisers Hyderabad       0.50      0.59      0.54        17

                   accuracy                           0.55       193
                  macro avg       0.54      0.58      0.55       193
               weighted avg       0.53      0.55      0.53       193
```

[46]:
```
new_match = pd.DataFrame({
    'team1': ['Chennai Super Kings'],
    'team2': ['Royal Challengers Bengaluru'],
    'toss_winner': ['Chennai Super Kings'],
    'toss_decision': ['bat'],
    'venue': ['M Chinnaswamy Stadium'],
    'match_type': ['League'],
    'season': [17],
    'target_runs': [220],
    'target_overs': [20],
    'super_over': ['N']
})
```

[45]:
```
predicted_winner = best_model.predict(new_match)
print("Predicted Winner:", predicted_winner[0])
```

```
Predicted Winner: Chennai Super Kings
```